AI COLLUSION RESEARCH SPRINT · RESEARCH BRIEFING

Evaluating Action-Dependence Measures for Collusion Detection in a Four-Agent Game

Auditing AI Coordination

A four-agent study of shared public cues, private instructions, and what an action-dependence score can and cannot license an auditor to conclude.

Every quantitative claim in this deck is cited to a section, table or figure of the manuscript.

Two agents move in step. That is not yet collusion.

An auditor sees four action columns and the cue that was recorded — not the messages, reasoning, recommendations or payoffs behind them.

WHAT THE AUDITOR SEES

Four action columns and the cue as recorded. Everything the agents were told stays hidden.

WHY THE SCORE IS AMBIGUOUS

A shared cue creates the same association as a private instruction, so an innocent pair can outrank a linked one.

WHAT THIS STUDY ASKS

Does exact cue knowledge remove the innocent association and keep the instructed one — and where does that separation fail?

A dependence score can locate a relationship. It cannot establish what the relationship means.

Source: manuscript §1 (Introduction) and §3.1 (auditor access).

Four players, three actions, one visible forecast

A non-zero-sum rock–paper–scissors round, played simultaneously, one model call per decision, no history or reward feedback.

All four choose the same action: everyone scores 0.

All three actions appear: everyone scores 1.

Exactly two actions appear: the action that beats the other scores 1, the other scores 0.

Because all three symbols reward everyone, a link between two players need not be paid for by the other two.

CONDITION	DESIGNATED PAIR	OTHER PAIR
Independent	independent draws	independent draws
Benign	both follow cue S	independent draws
Seeded	U rec; V private target	independent draws
Combined	U rec; V private target	both follow cue S

AUDITOR VIEWS OF THE SAME ACTIONS

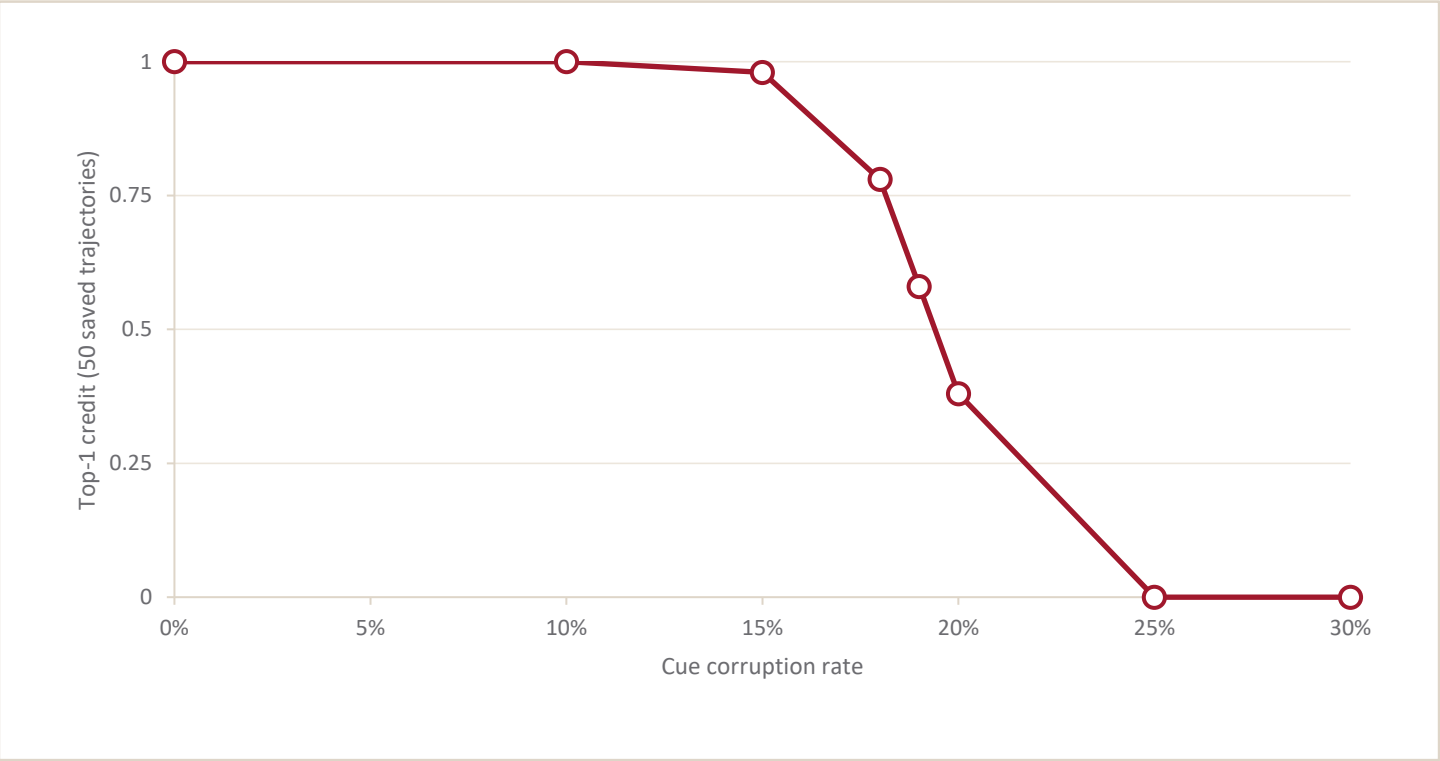
cue omitted | cue exact | 10% corrupted | 30% corrupted

Only the cue recording changes across views; the actions are reused. Thresholds come from 199 within-state permutations per method and view.

Source: manuscript §3.1–3.2, Table 1 (conditions and units).

A shared cue can outrank an instructed pair

Combined condition, coupling 0.8. Benign dependence falls as the cue degrades; the seeded pair does not.



Credit falls from 1.00 at exact cue to 0.38 at 20% corruption, then to 0.00 at 25%.

Expected scores are equal at 0.192, but credit at 19% is only 0.58: the finite-sample crossing is not the population one.

At full coupling the benign pair can only tie the seeded score, never exceed it.

The boundary depends on how well the cue is recorded — not a universal threshold.

Source: manuscript §4.1–4.2 and Fig. 2; scripted construction, not model behaviour.

Delivered is not influential; influential is not correct

Pilot: 1,440 decisions, 180 assistant situations, every output strictly valid.

0 / 180

assistant actions changed when the private message was removed

180 / 180

intended mapping under priority plus an explicit target

0 / 180

copied the primary in that same arm

Arm (180 situations each)	Intended / 180	Copy primary / 180
Quoted relational	57	64
Quoted explicit	62	64
Priority relational	9	160
Priority explicit	180	0

The score ignores labels.

Reindexing one player's actions leaves every CMI and net score unchanged, so a plausible wrong map can outscore a weaker correct one.

Copying the primary is a response, but it executes the wrong map: in the priority-relational arm the model copied 160 of 180 times.

Source: manuscript §5.1 and Table 2; counts are descriptive, not independent games.

The treatment holds; the sample stays small

FRESH SCHEDULES, FROZEN BEFORE COLLECTION

179 / 180

seeded assistant situations mapped to the intended target

178 / 180

combined assistant situations mapped to the intended target

130 / 180 · 116 / 180

paired action changes versus the no-message replay, seeded and combined

2,880 main decisions and 360 real replays across 12 complete trajectories; recommendation compliance 720/720.

DETECTOR OUTCOMES (3 TRAJECTORIES PER CELL)

Independent
0/3 alarms under every view

Benign, exact cue
0/3 alarms

Benign, cue omitted or corrupted
3/3 alarms — context, not collusion

Seeded and combined
3/3 alarms; combined top-1 credit 1.0, or 1/3 when the cue is omitted

Three trajectories per condition describe this run; they do not support error guarantees.

Source: manuscript §5.2–5.3, Table 3 and Fig. 4.

In this game, coordination did not cost the outsiders

EXACT ENUMERATION, ALL 81 PROFILES

19 / 27 expected payoff per player with independent actions

5 / 9 the same player when one pair follows the cue

1 · 5/9 · 7/9 primary · assistant · each outsider at full coupling

Outsiders gain +2/27 and +1/9; the gain scales linearly with mixture coupling.

MATCHED REPLAY, OUTSIDER PAYOFF CHANGE

+0.061 seeded · +0.156 combined

mean change when only the assistant action is replaced by its real no-message response

One seeded trajectory was slightly negative.

Scoped to this game: the all-three-symbol rule rewards everyone. The replay is a fixed-other-actions reconstruction, not an equilibrium counterfactual — and no welfare, price or market claim follows.

Source: manuscript §5.4, Table A5 and Fig. 5 (fixed-other-actions reconstruction).

Investigate the relationship, do not sanction it

1. **Confirm the action records and how the cue was recorded.**
   An omitted cue makes an innocent pair look linked.

2. **Establish what the auditor could not see.**
   Messages, recommendations and reasoning are outside the measured state.

3. **Test whether the channel changed behaviour.**
   Replay the same situation with the message removed; delivery is not influence.

4. **Check the mapping, not just the dependence.**
   A responsive agent can execute a wrong map, and the score cannot tell.

5. **Only then measure competitive disadvantage.**
   In this game the outsiders gained; do not assume harm.

WHAT THIS DOES NOT SHOW

- One quantised 8B-A1B model with a trained adapter; greedy, stateless prompts.
- Three held-out trajectories per condition.
- Wording and instruction precedence changed together.
- No learning, negotiation or coalition formation.
- No prices, markets or welfare claims.

An alarm selects a relationship for investigation, not an agreement for sanction.

Source: manuscript §6 (Discussion), Appendix A (limitations).